Programming for Mobile 4

Diogo Cabral
diogo.cabral@isel.pt

instituto superior de engenharia de lisboa

POLITÉCNICO DE LISBOA | POLYTECHNIC UNIVERSITY OF LISBON

Unity

There are two systems for creating runtime UI:

— Unity UI: Released with Unity 4.6 (late 2014)

— UI Toolkit: A new UI system that can be used alongside Unity UI

Unity

	Unity UI	UI Toolkit
What is it?	A GameObject-based UI system	A UI system inspired by web technologies
Workflow	UI elements are GameObjects inside a Canvas GameObject. They have a Rect Transform component to define position and scale relative to their parent.	UXML files define the UI's layout, and USS files define the styling. UXML files, also known as Visual Trees, are displayed in the Game view through a UI Document component applied to a GameObject.
Authoring tool	In the Scene view, create UI GameObjects in the Hierarchy and nest them to form layout groups.	With UI Builder, create UI Toolkit interfaces consisting of UXML and USS files visually.
Styling	Use Prefabs and Nested Prefabs to apply the same styling to many UI elements. You can override the Prefab's settings from the Scene view.	Use Selectors (USS files) to define styling and apply to many Visual Elements. Override styles with in-line styles in the Visual Tree elements.
Key benefits	Integration with other Unity systems, such as Animation, GameObjects, or the position in a 3D space	Performance (retained-mode graphics), scalability with complex UI, customizable materials, different screen sizes and theming; works for Editor UI as well as runtime UI; textureless interfaces
Best for…	Simpler UI elements blended in-game or with very specific needs, i.e., damage points coming from a character in a 3D space or a UI element that needs Physics	Recommended for screen overlay menus and HUD elements

Unity UI

Because the elements that make up each UI are GameObjects, they are compatible with other tools and systems within Unity. That said, all GameObjects must live within a Canvas GameObject for Unity UI.

The Canvas area is depicted as a rectangle in the Scene view. UI elements in the Canvas are drawn in the order they appear in the Hierarchy. The child element at the top renders first, the second child next, and so on. Drag the elements in the Hierarchy to reorder them.

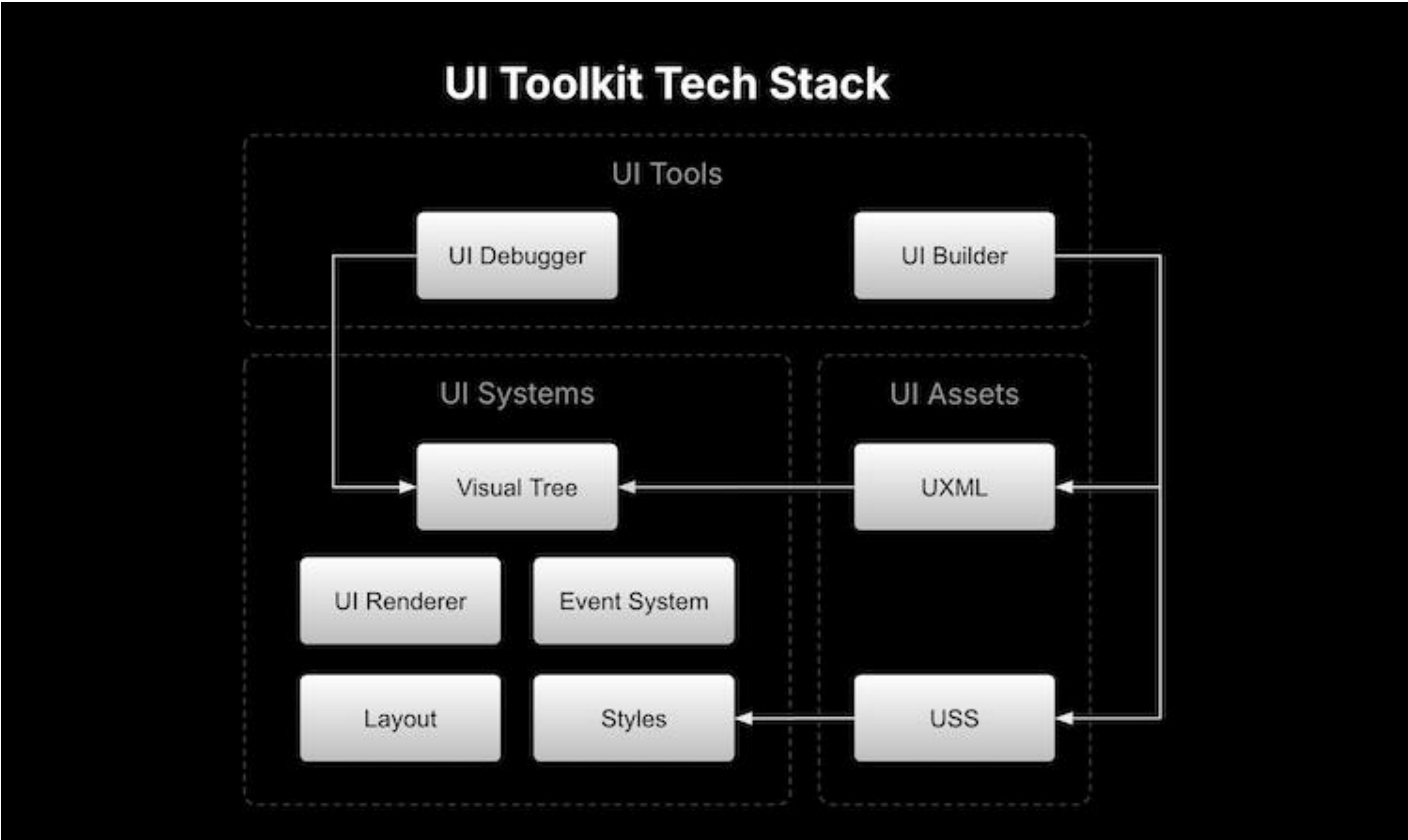
UI Toolkit

UI Toolkit to develop custom UI and extensions for the Unity Editor, and runtime UI for games and applications. UI Toolkit is designed for optimal performance and adaptability across platforms and it is inspired by standard ***web technologies***. UI Toolkit provides powerful tools and workflows that leverage existing knowledge of markup languages and style sheets.

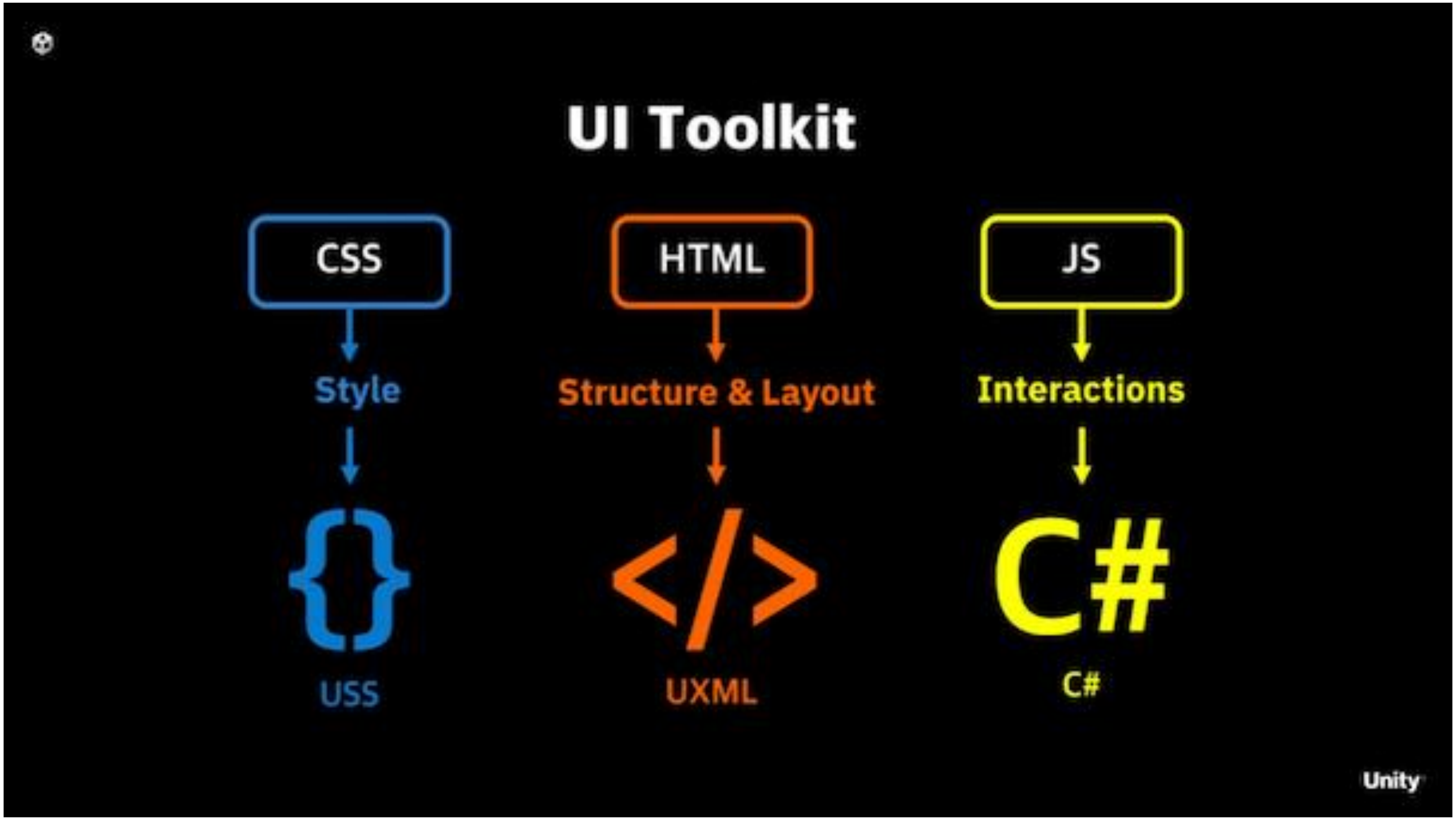
UI components are defined and managed as objects or nodes within a visual tree. Visual tree is an object graph made of lightweight nodes that holds all the elements in a window or panel. It defines every UI built with UI Toolkit.

https://docs.unity3d.com/6000.4/Documentation/Manual/ui-systems/introduction-ui-toolkit.html

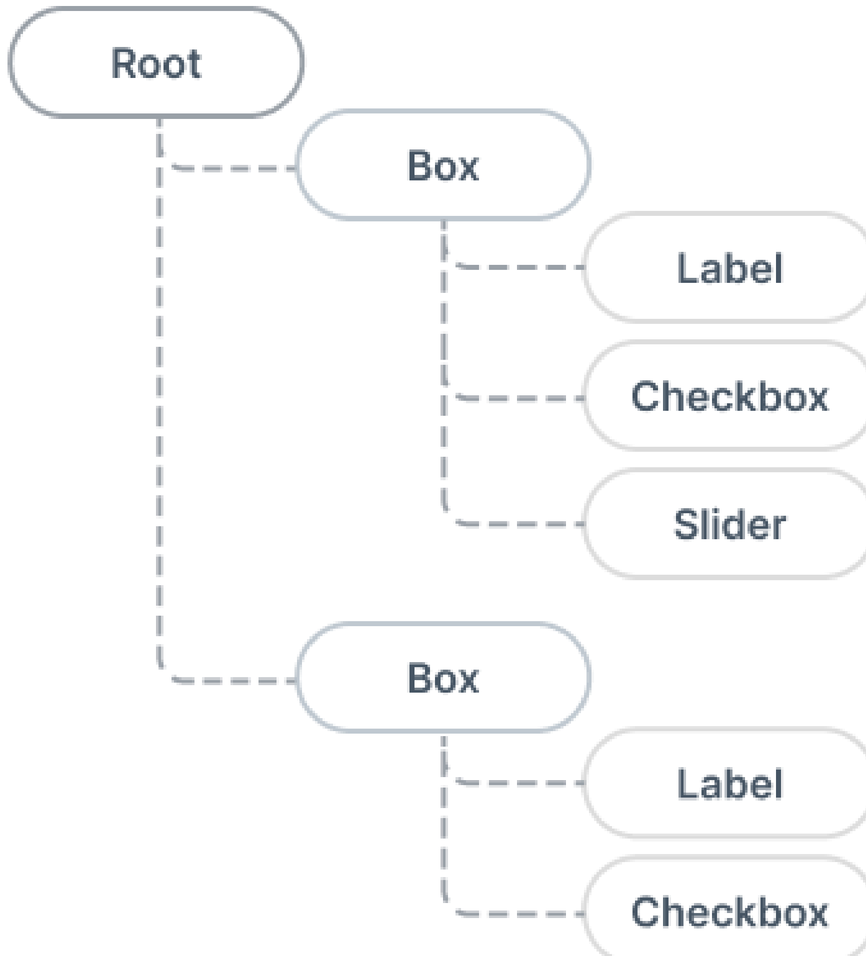
UI Toolkit


UI Toolkit Tech Stack
UI Tools
UI Debugger
UI Builder
UI Systems
UI Assets
Visual Tree
UXML
UI Renderer
Event System
Layout
Styles
USS

UI Toolkit


UI Toolkit
CSS
HTML
JS
Style
Structure & Layout
Interactions
USS
UXML
C#
Unity

UI Toolkit – Visual Tree



UI Toolkit

UI Editor

https://docs.unity3d.com/6000.4/Documentation/Manual/UIE-simple-ui-toolkit-workflow.html

UI on Game View (RunTimeUI)

https://docs.unity3d.com/6000.4/Documentation/Manual/UIE-get-started-with-runtime-ui.html

UI Toolkit

	SimpleCustomEditor	SimpleRuntimeUI
Base class	`EditorWindow`	`MonoBehaviour`
Runs in	Unity Editor only	Scene at runtime
UI entry point	`CreateGUI()`	`OnEnable()`
Gets root from	`rootVisualElement`	`UIDocument` component